Environmental Law Centre
UNIVERSITY OF VICTORIA

Murray & Anne Fraser Building
PO Box 2400 STN CSC
Victoria, BC V8W 3H7
Phone: 250.721.8188
Email: [illegible]
Web: [illegible]

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[Teck contact info]

Dear :

RE: Remediation of the former Sunro Mine at Jordan River

We write on behalf of the Juan de Fuca Salmon Restoration Society (JDFSRS), a non-profit society whose aim is to enable the rehabilitation and restoration of salmon habitat in creeks and rivers from the Sooke Basin to the San Juan River Valley. The JDFSRS is concerned with the copper contamination in the lower Jordan River from the former Sunro Mine site, which may interfere with the potential for habitat restoration in that waterway. We have identified Teck Resources Inc. (Teck) as a potential responsible person in relation to this contamination under British Columbia's *Environmental Management Act*, and we write to encourage Teck to proactively address this issue as soon as possible.

In the following submission we outline, to the best of our knowledge:

- the operating history of the site,
- the environmental values at stake in the area,
- the current evidence of contamination and its impact on fisheries restoration efforts,
- the current status of persons, including Teck, that may be identified as "responsible persons" by the BC Ministry of Environment under the *Environmental Management Act*, and
- the reasons why we believe Teck should consider undertaking remediation of the former Sunro Mine.

The Environmental Law Centre and the Juan de Fuca Salmon Restoration Society encourage you to use this information to ensure that the contamination at Jordan River is addressed in order to facilitate the reintroduction of salmon to this important fishery.

Site Description and Operating History

The Sunro Mine is located adjacent to the lower Jordan River at approximately 48°26'55" N, 124°01'55" W.[1] The mine was in production intermittently from 1962 to 1974, producing 1,465,017 tons of ore.[2] Copper was the primary mineral recovered, along with silver and gold.[3] It

[1] National Mineral Inventory, *Record 092C8 Cu1*, (Ottawa: Department of Energy, Mines and Resources, 1980), online: British Columbia MINFILE Database <http://www.em.gov.bc.ca/dl/PropertyFile/NMI/092C8_Cu1.pdf>. Title and tenure information found in the National Mineral Inventory was confirmed by reference to the Government of British Columbia's Mineral Titles Online and Tantalis GATOR services.

[2] *Ibid.*

further evidence comes to light, it may be difficult to show that Noranda Inc. or Xstrata Plc were "operators" and thus could be held liable as "responsible persons".

In contrast, as noted above, Consolidated Mining was directly involved in the creation of Sunro Mines Limited and was therefore apparently "responsible" for the initiation of the mine as that term is described in paragraph 111 of *Beazer*. Consolidated Mining was renamed Cominco in 1966. Cominco merged with Teck in 2001 to become Teck Cominco, which was renamed Teck Resources Ltd. in 2008. As discussed above, amalgamated corporations carry the liabilities of their predecessors. Teck Resources Ltd. should therefore be considered a "former operator" under the Act, and thus a responsible person. It does not appear to be exempt from that liability under section 46 of the EMA.[46]

The more recent claim-holders – Andris Kikuaka, Golden Sabre Resources Ltd., Christopher Billingsley, Kelly Funk, and Helmut Burke – may not be responsible for remediation of the site. First of all, as mere claim-holders, they may well not actually qualify as "owners" of the Sunro Mine site. Second, even if construed to be "owners" of the subsurface rights, they might well avoid responsibility for remediation of the site pursuant to s. 46 of the EMA and s. 22 of the CSR: their only interest would be a subsurface right, and we are not aware of evidence that they exercised that right in a manner that, in whole or in part, caused the site to become a contaminated site.[47]

The current and recent historical title holders would likely be captured as responsible persons under the EMA, section 45(1)(a) and (b). However, these owners would likely qualify for "minor contributor" status under section 50 of the EMA, which allows the Ministry to allocate a smaller portion of the remediation costs to responsible persons whose contribution to the contamination was small or non-existent. The evidence suggests that the contamination at the Sunro mine is primarily the result of its operations prior to 1974, and there is no evidence that the recent owners have added to or exacerbated this contamination.

In summary, the only apparent responsible person that we have been able to identify to date and that is not a minor contributor is Teck Resources Ltd.

While Teck may be the primary responsible person for the main mine site and associated tailings contamination, it should be noted that the log sort facility built by Western Forest Products from mine slump materials may have affected the productivity of the Jordan River watershed by infilling a salt marsh previously used by salmonids.[48] Teck may wish to work together with Western Forest Products to compensate for the combined impacts of their activities on the Jordan River salmon population.

[46] Section 29 of the Contaminated Sites Regulation, BC Reg 375/96 (CSR), provides that an owner is not exempt from liability simply because that owner did not cause the contamination directly, if that owner leased the property to another person and knew, or had a reasonable basis for knowing, that the lessee would cause the site to become contaminated. Considering the nature of mining operations generally, Consolidated Mining had a reasonable basis for knowing that its lessors would cause the site to become contaminated.

[47] EMA s 46(1)(n); CSR, s 22(1)(h) and 22(2).

[48] Burt, *supra* note 8.

> I also believe in using the word "responsible", the Legislature intended to include persons who brought about an operation in the sense of causing the operation to be carried on or carried out. Such a person would be responsible for the operation because, but for the actions or decision of that person, the operation would not have been carried on or carried out.[42]

In *Gehring et al v Chevron Canada et al*, the BC Supreme Court considered the application of the responsible person provisions of the EMA to dissolved corporations.[43] The court determined that a corporate person that has ceased to exist cannot be named as a responsible person.[44]

Corporations that are amalgamated with other corporations do not cease to exist, and the amalgamated corporation continues to be liable for the obligations of its predecessors, including obligations that did not exist at the time of amalgamation (for example, where companies were amalgamated before the EMA came into force).[45]

Analysis and application

In the case of the Sunro Mine, all corporate entities that directly owned or operated the mine during its operating period from 1962-1974 have now apparently been dissolved. The only two extant corporations connected to the mine are Teck Resources Inc. (the successor of The Consolidated Mining and Smelting Company of Canada Limited) and Xstrata Plc (the successor of Noranda Inc., which was the parent company of Noranda Exploration Company Limited). As noted above, Consolidated Mining and Noranda Exploration jointly owned Sunro Mines Limited, which owned the land on which the mine was located during the operating period and issued operating leases to other companies.

We submit that Consolidated Mining and Noranda Exploration were "operators" according to the definition in the Act. The formation of Sunro Mines Limited allowed for the consolidation of the Sunloch and Gabbro claims/grants and the subsequent issuance of operating leases, and the company was formed to carry out that purpose. This act of Consolidated Mining and Noranda Exploration thereby caused the operation to be carried on in the sense described in *Beazer* at paragraph 111.

The case of Xstrata is complicated. Xstrata Plc acquired Noranda Inc. (by then called Falconbridge Limited) in 2006, but Noranda Inc. was not the company that formed Sunro Mines Limited. Noranda had a subsidiary, Noranda Exploration Company Limited, that was involved in the Sunro deal. Although it is possible that Noranda Inc. exercised direct control over Noranda Exploration's transactions, we have not found evidence that that is the case. Similarly, we have not found evidence that Noranda Exploration was created specifically for the Sunro Mine. Unless

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[42] *Ibid* at para 111.

[43] 2006 BCSC 1639 at paras 52-55 & 76 [*Gehring*].

[44] *Ibid*. See also *British Columbia (Hydro and Power Authority) v. British Columbia (Environmental Appeal Board)*, 2003 BCCA 436, 17 B.C.L.R. (4th) 201 at paras 58-61, Newbury JA [*BC Hydro*], rev'd on other grounds 2005 SCC 1, [2003] 1 SCR 3.

[45] See *Canada Business Corporations Act*, RSC 1985 c C-44, s 186; *BC Hydro*, *supra* note 39, Rowles JA, dissenting, aff'd 2005 SCC 1, [2003] 1 SCR 3; R v Black & Decker Manufacturing Co. [1975] 1 SCR 411

In 2009 and 2010, sample EF04, taken immediately upstream of the tailings pile, had a total copper concentration of 3 µg/L, just above the applicable standard.[34] Sample EF02, from downstream of the tailings pile, had a total copper concentration of 16 µg/L in 2009 and 9 µg/L in 2010.[35] Prior to the fish flow release, observed concentrations at this downstream sampling point ranged from 40-90 µg/L.[36] Samples taken further upstream in 2009 and 2010 registered at or below the detection limit of 1 µg/L.[37]

BC Hydro has undertaken incubation studies in the lower Jordan River to begin evaluating the feasibility of reintroducing salmon, with promising results.[38] Recent studies suggest that BC Hydro's fish flow release experiment has been helpful in reducing the harmful effects of the contaminants through dilution.[39] However, the continuing exceedances suggest that copper still occurs in the river at levels that could interfere with fish productivity and condition. While containment of all seepage sources would be ideal, the removal or containment of the possible tailings pile adjacent to the river may allow copper concentrations in the river to return to near-natural levels, greatly lessening the strain on fish populations.

Identifying Responsible Persons

Legal framework

Persons defined as "responsible persons" may be ordered to remediate a contaminated site pursuant to section 48 of the *Environmental Management Act* (EMA),[40] or may be held liable under section 47 of the EMA for remediation costs incurred by a government body or another person. Paragraphs 45(1)(a) and (b) state that current and former owners and operators of a site are included in the definition of "responsible person".

The definition of "operator" in the EMA reads in part as follows: "…a person who is or was in control of or responsible for any operation located at a contaminated site…" This definition has been interpreted by the BC Supreme Court in *Beazer East Inc v BC (Environmental Appeal Board)*.[41] In that case, the BC Government named a company as a responsible person in respect of contamination caused by the operation of the company's subsidiary corporation. The remediation order at issue was made pursuant to the *Waste Management Act*, predecessor to the *Environmental Management Act*, but the definition of "operator" remains the same in the newer EMA. The court found that Beazer qualified as an "operator", and thus a responsible person, based on its control of and responsibility for the operation. Of the term "responsible", the court said the following:

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[34] *Ibid.*
[35] *Ibid.*
[36] *Ibid.*
[37] *Ibid.*
[38] Wright & Guimond, *supra* note 5; Guimond, *supra* note 24.
[39] Burt, *supra* note 8.
[40] [SBC 2003] c 53.
[41] 2000 BCSC 1698, 84 BCLR (3d) 88 [*Beazer*].

Logging operations have also been a feature of the Jordan River watershed since the 1880s, and at some point a log sort facility was constructed at the mouth of the river by filling in a salt marsh with materials from a mining slump.[26] Western Forest Products was the company responsible for this logging during the mine's operation.[27] Rights to the timber in this area were transferred in 2011 to the Pacheedaht Andersen Timber Holdings Limited Partnership.[28]

Environmental Values

Peak annual pink and chum salmon runs in the 1950's and 60's are estimated at 5,000-10,000.[29] The JDFSRS takes the position that the restoration of these historical numbers would provide a significant economic benefit to the residents of Jordan River and nearby communities with the development of a sport fishing industry. Additionally, reintroducing salmon in this river would provide an opportunity for local residents, including First Nations, to fish for food, social, and ceremonial purposes and for recreation. Jordan River is the traditional territory of the T'sou-ke First Nation and the Pacheedaht First Nation.

The Jordan River estuary, where most of the mine tailings were deposited, is a popular surfing destination. There is no information available on the sediment quality on the beach, its potential for health impacts on beach users, or its potential to contaminate shellfish and other seafood in the area. Further investigation may be warranted.

Evidence of Contamination

Water in the lower Jordan River was sampled for copper contamination and other characteristics annually from 2005 to 2010 as part of BC Hydro's Jordan River Water Use Plan.[30] Copper concentrations downstream of the possible tailings pile that abuts the river have consistently exceeded the water quality standards for the protection of aquatic life listed in Schedule 6 of the Contaminated Sites Regulation (CSR).[31] The sample results show a marked decrease in copper concentrations coinciding with flooding in 2006-2007 and the beginning of the fish flow release from the Elliot Dam in 2008.[32] However, copper concentrations downstream of the mine remain above the CSR standard.[33]

As the samples were taken from within a watercourse that supports fish, the assumption of ten times dilution in footnote 2(a) of Schedule 6 of the CSR does not apply. The water falls into the lowest hardness category for aquatic life standards for copper. The applicable standard for this watercourse is therefore 2 µg/L.

Feasibility Study, (2005), online: BC Hydro <http://www.bchydro.com/bcrp/projects/docs/vancouver_island/04Jo01.pdf>.

[26] Wright & Guimond, *supra* note 5.

[27] *Ibid.*

[28] Burt, *supra* note 8.

[29] Wright & Guimond, *supra* note 5.

[30] Burt, *supra* note 8.

[31] *Ibid.*

[32] *Ibid.*

[33] *Ibid.*

After the Crown grants were surrendered in 1996, a four-post claim was staked by Andris Arturs Kikuaka in 2003.[14] This claim was converted to a cell claim in 2005 and transferred to Golden Sabre Resources Ltd. in 2006.[15] In 2007, the claim expired and a new one was staked by Christopher John Billingsley.[16] Kelly Brent Funk acquired that claim in 2010 and staked several more claims from 2009 through 2012.[17] An additional claim at the location of the former camp and portal was staked by Helmut Burke in 2011.[18] All claims are currently in good standing, but are being maintained with fees paid in lieu of exploration and development work.[19]

In addition to the active mining claims, there are several surveyed parcels covering the area of the former mine and held in fee simple. These parcels have changed hands several times over the last few years.[20] All of the following lots are in the Renfrew District. District Lots 922 and 802 cover the core area of the mine, while the other lots listed surround them to the south and west.

- District Lot 922 is currently owned by Sun Prairie Holdings Inc. and was previously owned by Western Forest Products from 1997 to 2010 (initially Western Forest Products Limited and later Western Forest Products Inc.).
- District Lot 802 is owned by Donald Keith Steffler, and has previously been owned by Evelyne Reine Gauthier, Chalis Design Limited, Brotherston Logging Co. Ltd., and Western Forest Products Inc.
- District Lot 532 and Section 3 are registered to Canadian Puget Sound Lumber and Timber Company Limited, but that company no longer exists.
- District Lot 725 is owned by Peter Wilson Bailey, and has previously been owned by A-3 Holdings Corp and Western Forest Products Inc. and Western Forest Products Limited.
- Section 7 Remainder is owned by Western Forest Products Inc.

The Jordan River is also home to hydroelectric generation facilities, with the first constructed in 1909.[21] Beginning in 1971, most of the river volume was diverted through a penstock, leaving a long stretch of the riverbed with greatly reduced flows.[22] The various alterations to the river made by BC Hydro significantly reduced the availability of habitat for anadromous fish in the lower Jordan River.[23] BC Hydro is experimenting with allowing greater releases from the Elliot Dam to maintain water levels in the lower Jordan that can support fish, and fish and habitat monitoring has shown marked improvements since the release began in 2008.[24] However, copper contamination still remains a major barrier to the reestablishment of self-sustaining salmon populations.[25]

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[14] Mineral Titles Online, tenure 404235, online: <www.mtonline.gov.bc.ca>.
[15] *Ibid.*
[16] *Ibid*, tenures 564267, 663263, 840877, 840887, 841971, 842841, 887509, 889694, 953676, and 963309.
[17] *Ibid*, tenure 564267
[18] *Ibid*, tenure 897708.
[19] Mineral Titles Online, *supra* notes 17 & 18.
[20] BC Land Titles Registry.
[21] Wright & Guimond, *supra* note 5.
[22] *Ibid.*
[23] *Ibid.*
[24] Burt, *supra* note 8.
[25] DW Burt, *Lower Jordan River Fish Index Study – Year 5 Results (2009)*, (2010), online: BC Hydro <http://www.bchydro.com/about/sustainability/conservation/water_use_planning/vancouver_island/jordan_river.html>. See also Wright & Guimond, *supra* note 5; E Guimond, Jordan River Pink Salmon Transplant

was an underground mine with a mill constructed in a cavern below the surface.[4] Ore was processed underground and tailings were piped out to the ocean.[5] Pipeline breaches are known to have caused occasional releases of tailings along the route.[6] Tailings may also have been deposited on the bank of the river, and this tailings pile occasionally slumps into the watercourse.[7] There is, however, some debate over whether this pile is tailings or merely waste rock from the excavation of the mine portal.[8] In December 1963, a collapse allowed the Jordan River to flow into the mine.[9] The flooded chamber was used as a water supply during later operation, and portions of the mine remain flooded.[10] Mine shafts and seepages continue to contaminate the Jordan River with copper.[11]

The Sunro Mine began as two separate groups of adjacent claims: Sunloch and Gabbro. Both were staked in 1915. The first of the Sunloch claims were Crown-granted to The Sunloch Mines Limited, George Winkler, Christian Frank, and David Hanbury in 1918, and additional claims were granted in 1924-25. In 1919, The Consolidated Mining and Smelting Company of Canada Limited ("Consolidated Mining", renamed Cominco in 1966) purchased a controlling share in The Sunloch Mines Limited. The Gabbro claims were Crown-granted to The Gabbro Copper Mines Limited first in 1921, with additional grants in 1924. Miscellaneous exploration activities took place through the 1920s, 40s, and 50s. In 1955, Noranda Exploration Company Limited acquired a controlling share of The Gabbro Copper Mines Limited. In 1956, Noranda and Consolidated Mining formed a new company, Sunro Mines Limited. Ownership of all the lots was transferred to this new company in 1957, and it retained ownership until the Crown grants were surrendered in 1993.[12]

In 1960, Cowichan Copper Co. Ltd. (later called Cerna Copper Mines Limited) obtained an operating lease from Sunro Mines Limited and began work on the underground mill, which went into operation in 1962. Cowichan operated the mine intermittently from 1962 until 1968, with the help of capital investment in 1965-66 from Sheep Creek Mines Limited (later called Aetna Investment Corporation Limited). In 1969, Crownex International Ltd. acquired a new operating lease from Sunro Mines Limited. In 1970, a subsidiary of Crownex began underground development before Crownex subleased the property to Pechiney Development Limited. In 1971, Pechiney and Crownex jointly incorporated Jordan River Mines Limited, which operated the mine until it closed for the last time in 1974. Before the mine closed, Pechiney sold its controlling interest in Jordan River Mines to Crownex (by then called Dison International Ltd.)[13]

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[3] *Ibid.*
[4] *Ibid.*
[5] MC Wright & Esther Guimond, *Jordan River Pink Salmon Incubation Study*, (2003), online: BC Hydro <http://www.bchydro.com/bcrp/projects/docs/vancouver_island/02JO64.pdf>.
[6] *Ibid.*
[7] *Ibid.*
[8] DW Burt, *Summary of Water Quality and Biological Data for Anadromous Reaches of the Jordan River*, (2012) [unpublished draft prepared for the BC Hydro Fish and Wildlife Compensation Program] at 24.
[9] Wright & Guimond, *supra* note 5.
[10] Personal communication with Randy Clarkston.
[11] Wright & Guimond, *supra* note 5.
[12] National Mineral Inventory, *supra* note 1; BC Land Titles Registry.
[13] National Mineral Inventory, *supra* note 1.

Policy Basis for Remediation of the Sunro Mine

The Jordan River has been decimated by decades of diversion and contamination. Nevertheless, it has significant potential to offer high environmental values to the region. The reintroduction of salmon to this watercourse would provide a source of food for local First Nations and other residents, and would provide a significant economic opportunity for wilderness tourism.

The environmental management regime in British Columbia is premised on the polluter pays principle. The predecessor to Teck Resources Ltd. apparently contributed to the cause of the contamination at the Sunro Mine, and profited from the activity that created it. As the last remaining corporate entity with a direct connection to the former mine, Teck has a responsibility to remediate the contamination that continues to interfere with the reestablishment of salmon stocks in the river.

This responsibility extends beyond legal rules, however. We urge Teck to set an example of corporate social responsibility by giving back to the community of Jordan River that supported its enterprises in the past. The legacy of the Sunro Mine should and can be a positive one. We therefore encourage Teck to take immediate action to address the remaining contamination that threatens the Jordan River, and help to once again make the river a rich ecosystem capable of supporting salmon and other species.

Thank you for your consideration. Should you have any questions, please contact the undersigned.

Sincerely,

Matthew Nefstead
Student, Environmental Law Centre

Calvin Sandborn
Lawyer and Legal Director, Environmental Law Centre